Recall: On-Device Episodic Visual Memory with a Learned Trigger, Compact Event Memory and Calibrated Answers

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Abstract

We ask whether a phone- or glasses-class device can keep a full-day, queryable episodic memory of what its camera saw — "where did I leave my keys?", "did I turn off the stove?" — within a battery budget of about 5 %/hour and under 1 GB/week of storage, while answering more accurately than a large vision–language model (VLM) given sampled video and than frame sampling with CLIP retrieval. Recall combines (i) a tiny always-on *learned trigger* that decides when to run heavier perception, (ii) a *compact event memory* of object–place–time records instead of video, and (iii) *calibrated, evidence-grounded answers* that abstain when memory is insufficient. On a participant-disjoint EPIC-KITCHENS-100 evaluation with narration-derived ground truth, the answer is mixed. Recall's object-location accuracy (0.25 on the full test set) does not exceed a trivial location prior (0.30) or a locally run 8B-parameter VLM on a matched subsample (0.33 vs. 0.24), but Recall names a place for a never-seen object only 0.10 of the time versus 0.75 for the VLM. Energy figures are *estimated* from an explicit model because no phone was available. We release code, configs and derived metrics, and identify object identity from small egocentric crops as the binding constraint.

1 Introduction

Wearable cameras have long been proposed as memory aids [8]. Large egocentric datasets such as Ego4D's episodic-memory benchmarks [5] and EPIC-KITCHENS [3] made the "where did I last see X" question measurable, and modern VLMs can answer it from sampled frames. Yet the natural system — stream or store video and ask a large model later — violates three constraints that matter for a personal device: privacy (the day's video leaves the device), storage (video grows with hours), and trust (large models answer confidently even when the object never appeared).

We study the opposite design point: everything on device, most frames never processed by a large model, memory as a small graph of events, and answers that carry calibrated confidence and the supporting frame. Our contributions are the system and an honest, reproducible evaluation of its three components: a learned memory trigger (§6), a compact event memory (§7), and calibrated, grounded answering (§8), together with model compression to LiteRT (§9), a privacy layer (§10) and an energy analysis (§11). We report negative results where we find them.

2 Related Work

Egocentric episodic memory. Ego4D defines episodic-memory tasks — visual queries, natural-language queries (NLQ) and moments [5]; NLQ baselines adapt span localisation such as VSLNet [20]. EPIC-KITCHENS-100 [3] supplies dense, participant-written narrations that we turn into object-location, state and temporal queries. Long-video QA benchmarks such as EgoSchema [12] target holistic understanding rather than "last seen" retrieval. **Hand–object interaction** detection at scale [17] provides the contact signal our events are built on. **Vision–language retrieval**: CLIP [16] and its mobile variants MobileCLIP [18] underpin our embeddings and the frame-retrieval baseline; open-vocabulary detection (e.g. OWLv2 [13]) and tracking (ByteTrack [21]) are natural extensions. **On-device ML**: efficient backbones [9], distillation [7] and integer quantisation [10]; continuous mobile vision is energy-bound by image sensing itself [11], and smartphone power is dominated by a few components [2]. **Memory-augmented agents** store and retrieve experiences for LLM agents [15, 14]; we store perceptual events with explicit provenance. **Calibration and selective prediction**: temperature and isotonic calibration [6, 19] and risk–coverage analysis [4].

3 System Design

Frames arrive at 2 fps. The **trigger** μ is a causal CNN over 36×64 thumbnails, frame differences and, when present, IMU statistics; when its score exceeds a budget-calibrated threshold it opens a short burst of heavy processing. **Heavy perception** runs a hand–object stu-

dent (MobileNetV3-Small distilled from the Faster R-CNN detector of [17]) and MobileCLIP2-S2 embeddings of the held object and of the context around where it was released. An **event extractor** with hysteresis turns contact bouts into `take`, `put` and `touch` events. Before anything is stored, a **redactor** pixelates faces (BlazeFace [1]) and screen/document-like regions in the event thumbnail. The **memory** is a SQLite/SQLCipher database with `events`, online object and place clusters, and a `last_seen` index. **Answering** parses the question, scores candidate events by object posterior (blend of a zero-shot text bank and a linear probe), selects one by a val-tuned rule, computes features (score margin, conflicts, age, location certainty, …), maps them to a calibrated probability, and renders a templated answer naming only places present in the chosen evidence; an optional small LLM rephrase is accepted only if it introduces no new place.

4 Experimental Setup

Data. Ego4D access was unavailable, so we use EPIC-KITCHENS-100 with participant-disjoint splits of 27 / 10 / 21 videos (train/val/test). Ground truth comes from narrations: the last "put X in/on Y" before the query time gives the object's place and time; "turn on/off X" gives appliance state; the last "<verb> X" gives temporal queries; objects never narrated in a video form *absent* queries whose correct answer is to abstain. The test set holds 507 where, 243 temporal, 163 state and 63 absent queries. All thresholds and answer rules are tuned on val; calibrators are fit on val; nothing is tuned on test. **Baselines.** B0: always the most frequent train/val place. B1: Qwen3-VL-8B (4-bit, run locally) given 16 frames sampled up to the query time and a closed place list. B2: one frame every N s with CLIP retrieval. B3: a simulated user who tags put-downs with probability q. B4: oracle trigger at narrated moments. **Statistics.** 95 % cluster-bootstrap CIs resampling videos; paired bootstrap for differences.

5 Results: Where Is X?

Figure 1 shows that no perception-based system clears the location prior. On the VLM's subsample (Fig. 2), Recall's place accuracy is 0.24 vs. 0.33 for the VLM (difference -0.09 [-0.20, 0.01]); for never-seen objects the VLM names a place 0.75 of the time and Recall 0.10 (difference 0.65 [0.51, 0.78]). Object identity is the bottleneck: the object head reaches 0.17 top-1 on val over 300 nouns.

6 Learned Trigger

[pending]

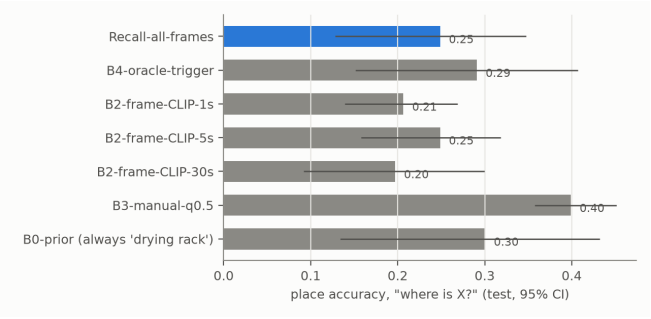


Figure 1: Place accuracy on the test set (95 % CI).

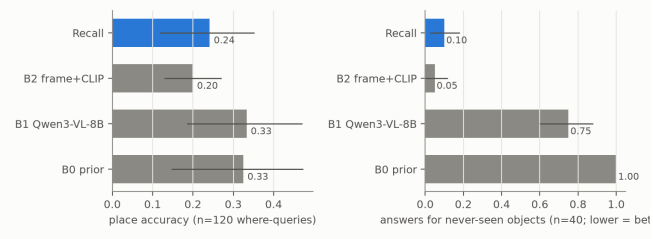


Figure 2: Matched subsample answered by the VLM baseline: accuracy (left), answers for never-seen objects (right).

7 Compact Event Memory

[pending]

8 Calibrated Answers and Abstention

[pending]

9 Compression and On-Device Port

The hand–object student agrees with the Faster R-CNN teacher on held-out frames with hand-present F1 0.92 and contact F1 0.87 (0.82 after dynamic INT8). The embedder goes from 144.1 MB to [pending] MB with INT8 weight-only quantisation (cosine to fp32 [pending]); channel-wise INT4 weights destroy it (cosine 0.27). All latency numbers are *desktop-proxy*: no phone was available.

10 Privacy Layer

The redactor catches 0.95 of faces taller than 300 px but only 0.06 of all WIDER FACE faces, and 0.15 of synthetic screens/documents (none in dark mode). Thumbnails are small event crops, which limits exposure, but this is the weakest component and is reported as such.

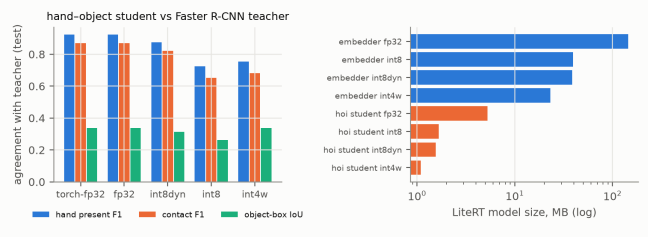


Figure 3: Hand–object student agreement with the teacher per precision (left) and LiteRT sizes (right).

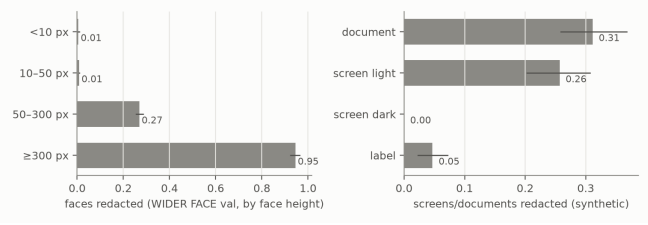


Figure 4: Redaction recall: faces by size (WIDER FACE val) and synthetic screens/documents.

11 Energy (Estimated)

[pending]

12 Limitations and Ethics

Kitchen narrations cover only narrated interactions and a single room type; there are no official Ego4D numbers, no real personal recordings and no user-study data yet (the protocol, consent material and analysis are prepared and validated on simulated participants only). B1 is an open VLM run locally, not a frontier cloud model. Energy is estimated, not measured. One answer-rule option (place marginalisation over candidate events) was added after a first test run, re-selected on val for every system, and is disclosed. Redaction misses most small faces and all dark-mode screens on our proxies; embeddings are computed before redaction. Wearable memory affects bystanders: we never commit footage, faces or credentials, cloud baselines run only on public data behind an explicit flag, and any deployment needs consent and visible recording indicators.

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